

Detailed Notes: Text Generation and AI Summarization

1. Components of Text Generation

Text generation in Generative AI involves several stages: **Input Prompt:** User provides instructions or a query. **Tokenization:** Text is converted into smaller units called tokens. **Language Model Processing:** The model analyzes token relationships using learned patterns. **Context Prediction:** The model predicts the most probable next words based on context. **Response Generation:** Tokens are combined into meaningful text output.

2. Gemini API Workflow

The screenshots demonstrate how to use Google's Gemini API: Install the library using **pip install google-generativeai**. Import the package in Python. Configure the API using an API key. Load a Gemini model such as Gemini 1.5 Flash. Send a prompt to the model. Receive and display the generated response.

3. Model Processing Step

After tokenization, the model processes the tokens and analyzes context. It determines semantic relationships, grammar, intent, and relevant information before generating output.

4. Next Word Prediction

Large Language Models generate text by repeatedly predicting the next most likely token. Example: Input: "Dear Sir" Predicted continuation: appropriate formal sentence structure.

5. Response Generation Example

Example output shown in the screenshots: Dear Sir, I am not feeling well and would like to request leave for one day. Regards, Shreya This demonstrates how an AI model converts a short prompt into a complete professional message.

6. AI Document Summarization

Summarization is the process of reducing lengthy text while preserving key information. **Key Benefits of AI Summarization** Converts long documents into shorter summaries. Retains important information. Improves readability and understanding. Saves time during content analysis. Helps users quickly identify important points. Useful in research, education, business, and content management.

7. Practical Applications Email drafting and professional communication. Chatbots and virtual assistants. Content creation and blogging. Document summarization. Question answering systems. Customer support automation. Educational tools and tutoring systems.

8. Interview/Exam Notes Token = smallest unit processed by an LLM. Prompt = instruction given to the model. Context helps maintain coherence in responses. LLMs generate text token-by-token. Summarization can be extractive or abstractive. Gemini is a multimodal AI model family developed by Google.